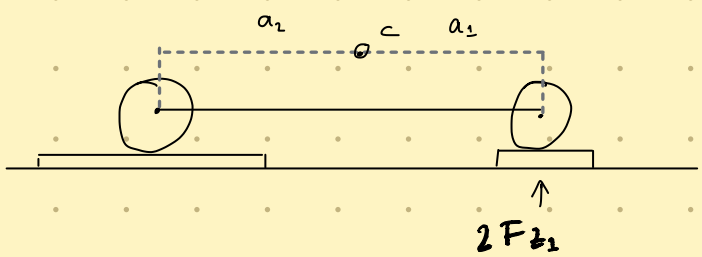
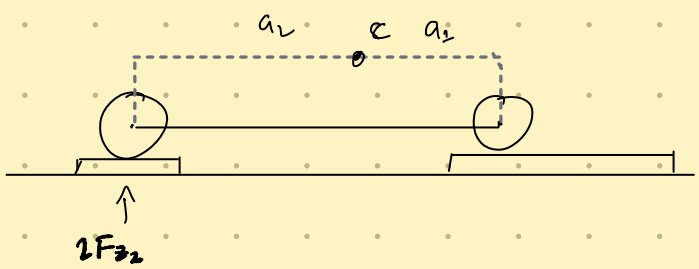
NoRa9

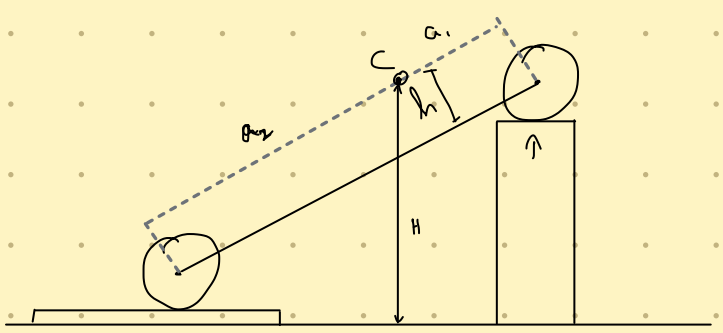
Front axle weight force.



Rear axle weight force:



Mass height center determination:



Total weight = 300.5 kg

Wheel base 'l' = 2.5 m

Radius of tire 'R' = 468 mm = 0.468 m

$$2F_{z_1} = 2mg = 2 \times 77.25 \times 9.8$$

$$2F_{z_1} = 1.514.1 \text{ N}$$

$$2F_{z_2} = 2mg = 2 \times 73 \times 9.8$$

$$2F_{z_2} = 1430.8 \text{ N}$$

Mass centre position: a_1, a_2

$$a_1 = \frac{2l}{mg} F_{z_2} = \frac{2.5}{300.5 \times 9.8} \times 1430.8 = 1.214$$

$$a_1 \simeq 1.25 \text{ m}$$

$$a_2 = \frac{2l}{mg} F_{z_1} = \frac{2.5}{300.5 \times 9.8} \times 1.514.1 = 1.285$$

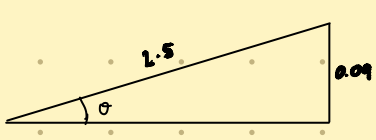
$$a_2 \simeq 1.3 \text{ m}$$

Vertical position 'h' of centre 'c' is,

$$h = R - \frac{a_1 F_{z_1} - a_2 F_{z_2}}{0.5\, mg} \cot\theta$$

$$h = R + \left(a_2 - \frac{2F_{z_1}}{mg} l\right) \cot\theta$$

Observation 1:



$$2F_{z_1} = 2mg = 153.5 \times 9.8$$

$$2F_{z_1} = 1504.3 \text{ N}$$

$$\sin\theta = \frac{0.09}{2.5} = 2.06$$

$$\theta = \sin^{-1}(0.036)$$

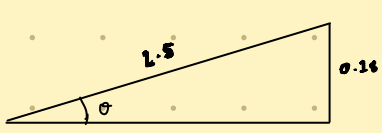
$$\theta = 2.0631^\circ$$

$$h = 0.468\text{m} + \left[1.3 - \left(\frac{1504.5}{300.5 \times 9.8}\right) 2.5\right] \cot(2.063)$$

$$= 0.468 + [0.0228](-0.536) = 0.468 + (-0.0222)$$

$$h = 0.456 \text{ m} = 45.6 \text{ cm}$$

Observation 2:



$$2F_{z_1} = 2mg = 171.5 \times 9.8$$

$$2F_{z_1} = 1680.7 \text{ N}$$

$$\sin\theta = \frac{0.18}{2.5}$$

$$\theta = \sin^{-1}(0.072)$$

$$\theta = 4.1288^\circ$$

$$h = 0.468 + \left[1.3 - \left(\frac{1680.7}{300.5 \times 9.8}\right) 2.5\right] \cot(4.1288)$$

$$= 0.468 + [-0.1267](0.6603) =$$

$$= 0.468 - 0.0836$$

$$h = 0.384 \text{ m} = 38.4 \text{ cm}$$

Raguvandhan